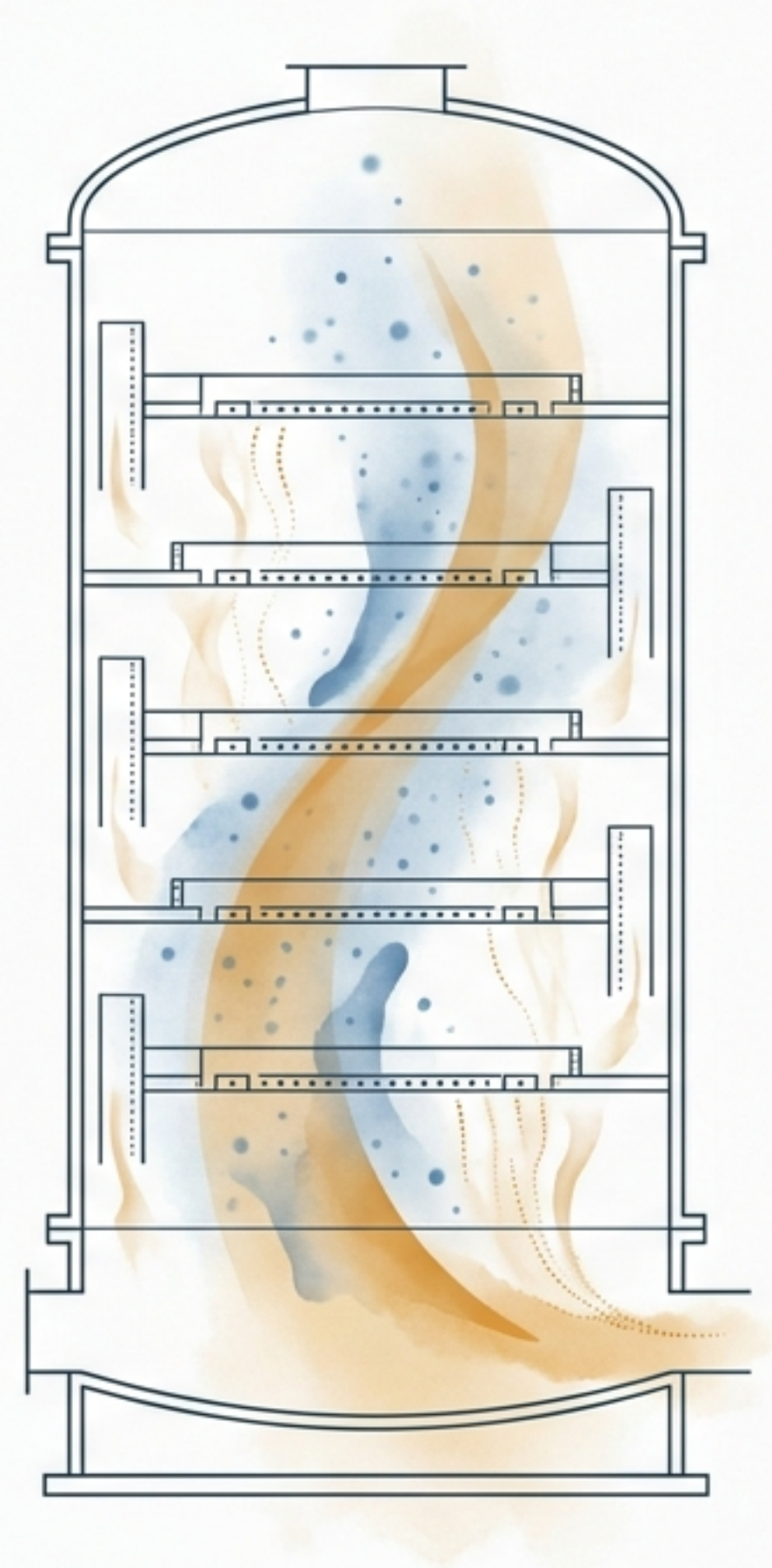


The Architecture of Dehydration

Sizing Inlet Scrubbers & Glycol Contactors

Prepared by Engineer Sajjad k. Hano



Thinking in Reverse: The Anatomy of System Failure

UNDERSIZED VESSEL

Symptom: Fluid Carryover & Foaming

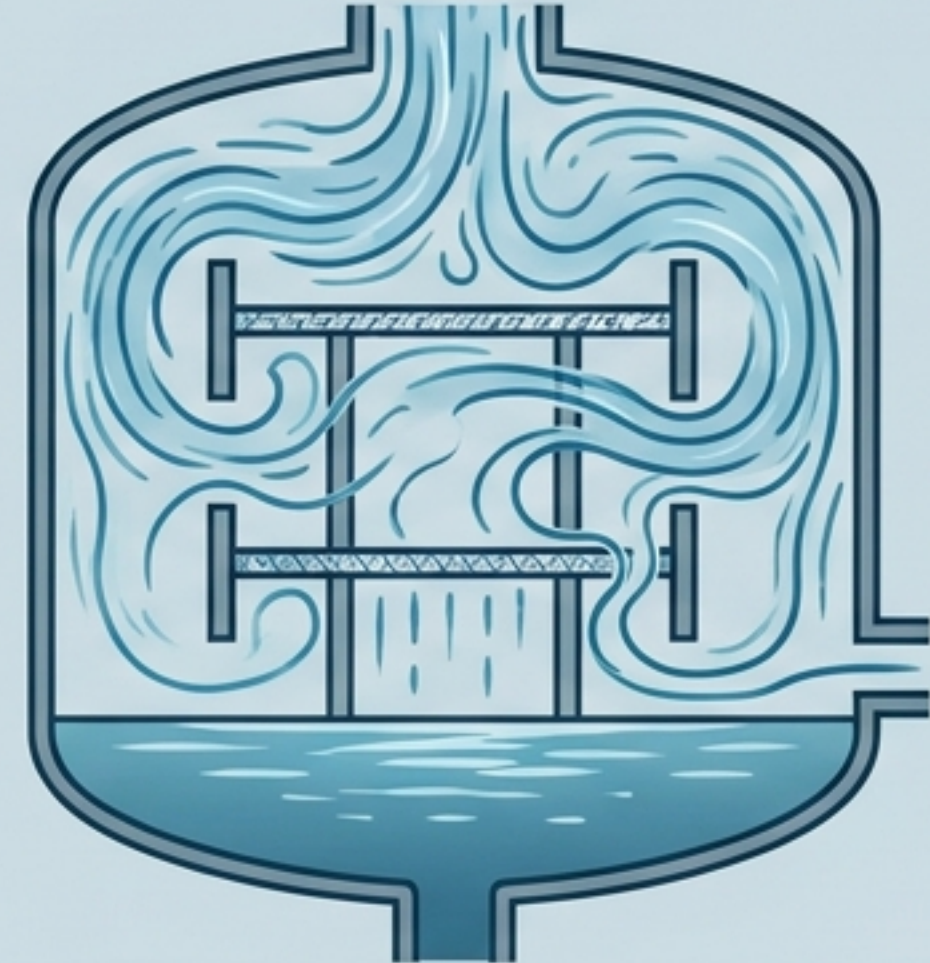
Consequence: High vapor velocities overwhelm droplet settling limits, leading to catastrophic glycol loss and premature flooding.



OVERSIZED VESSEL

Symptom: Channeling & Wasted CAPEX

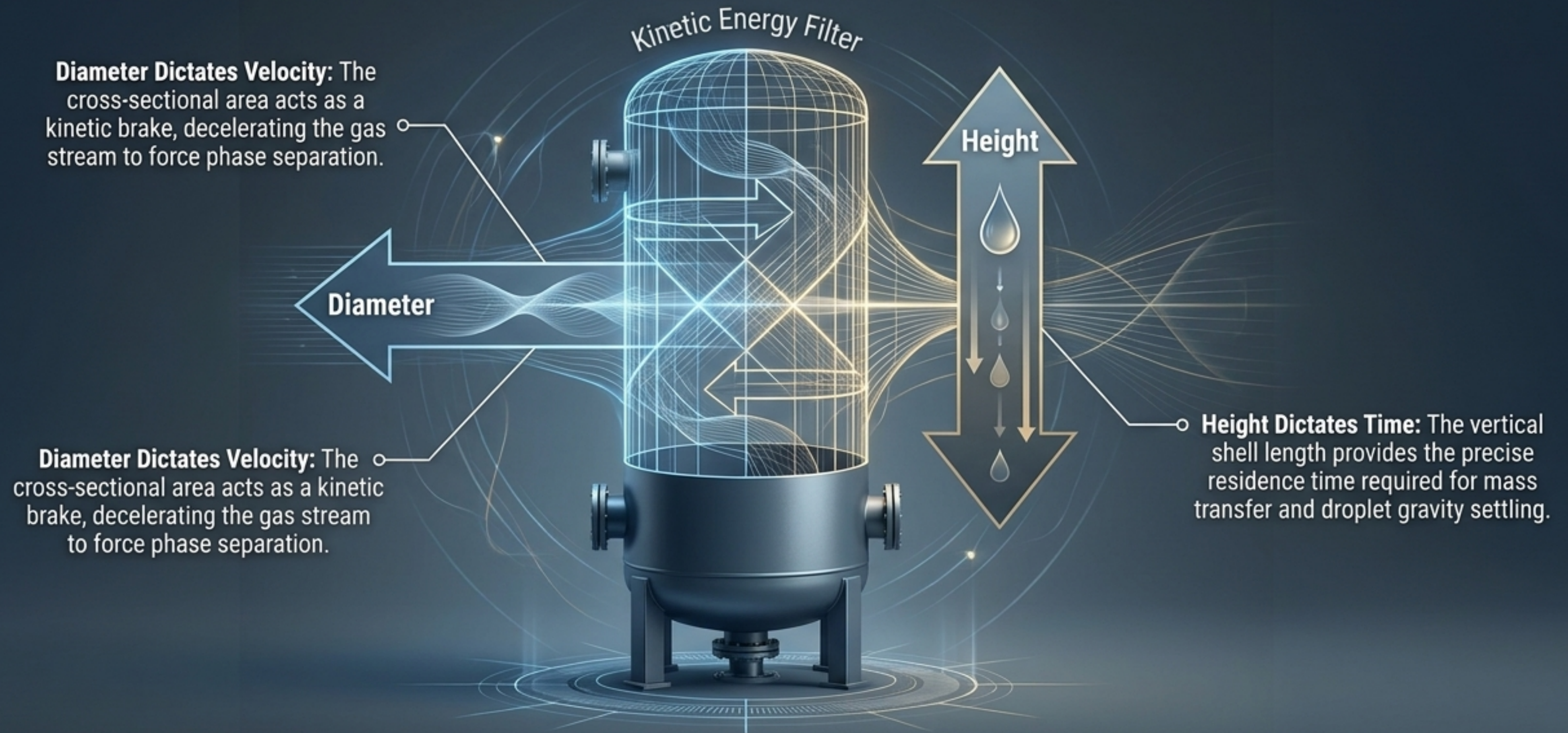
Consequence: Vapor bypasses liquid contact zones. The system fails to meet the required dew point despite excessive physical footprint.



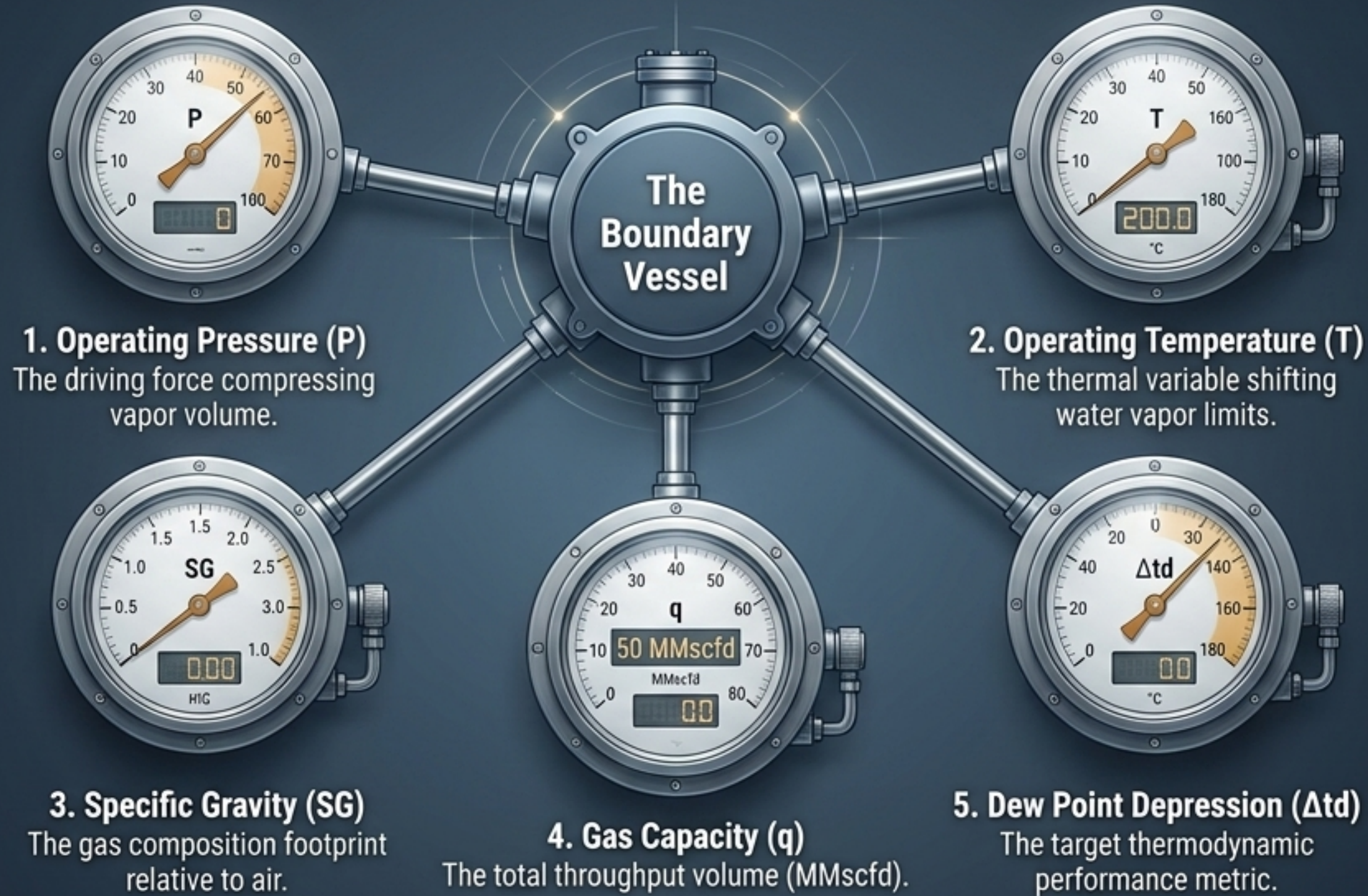
"The Sivalls (1977) empirical charts are not mere suggestions; they are the exact mathematical boundaries separating thermodynamic equilibrium from operational chaos."

Unconventional Framework: Sculpting Kinetic Energy

We traditionally view scrubbers and contactors as static containers holding fluids. To master sizing, we must view them as active energy manipulators. We are not sizing steel plates; we are sculpting gas flow geometries to force dehydration.



Extracting the Core Variables



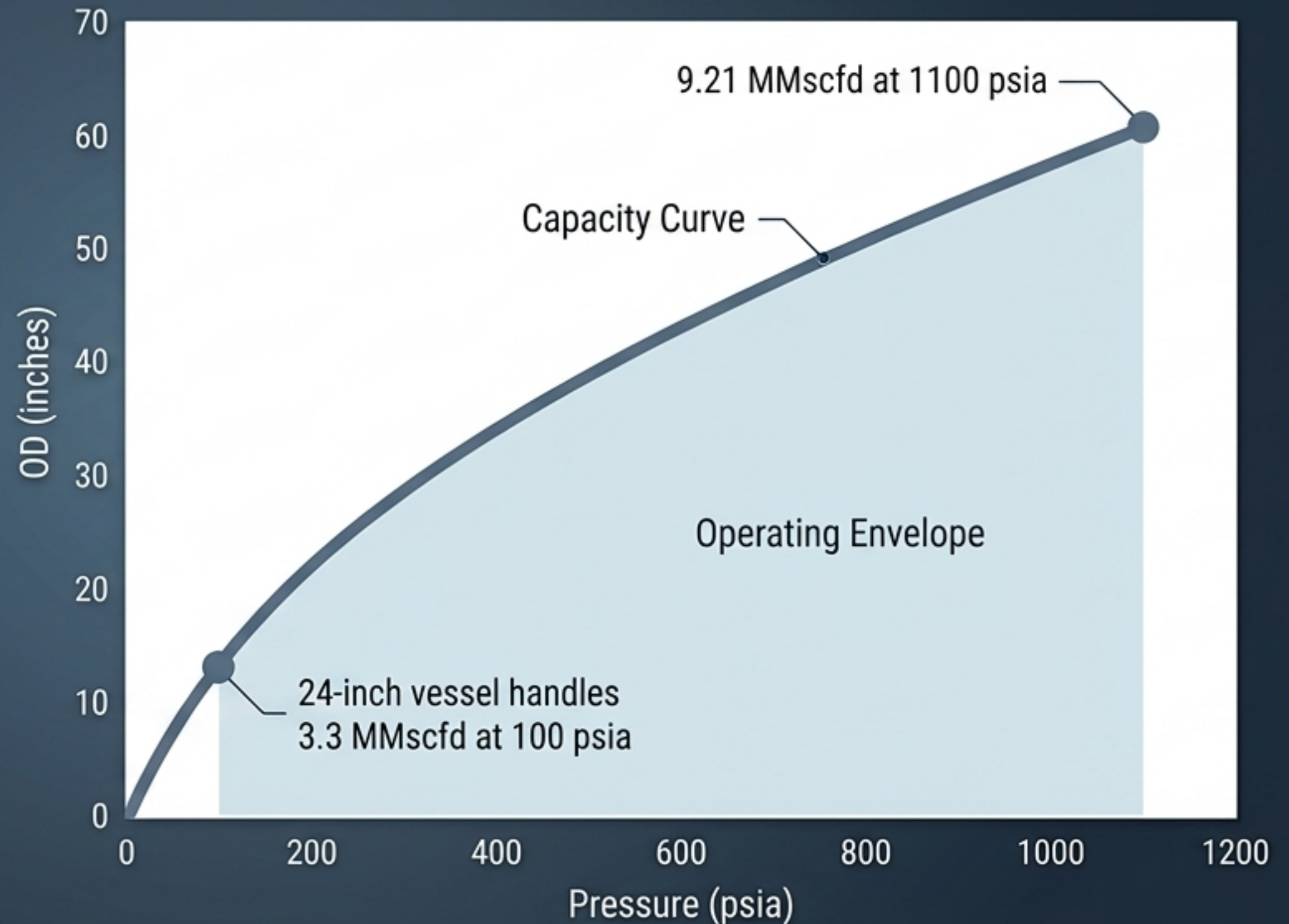
Every lookup table, correction factor, and structural dimension in the Sivalls methodology is governed by the triangulation of these five kinetic inputs.

The First Line of Defense: Vertical Inlet Scrubbers

The Baseline: Sivalls (1977) sizing is anchored at 0.7 SG and 100°F.

The Envelope: Capacity jumps non-linearly as pressure increases. A 24-inch vessel handles 3.3 MMscfd at 100 psia, but expands to handle 9.21 MMscfd at 1100 psia.

Insight: Pressure is a volumetric equalizer. Higher pressure shrinks the actual gas volume, allowing massive throughput in surprisingly small structural diameters.

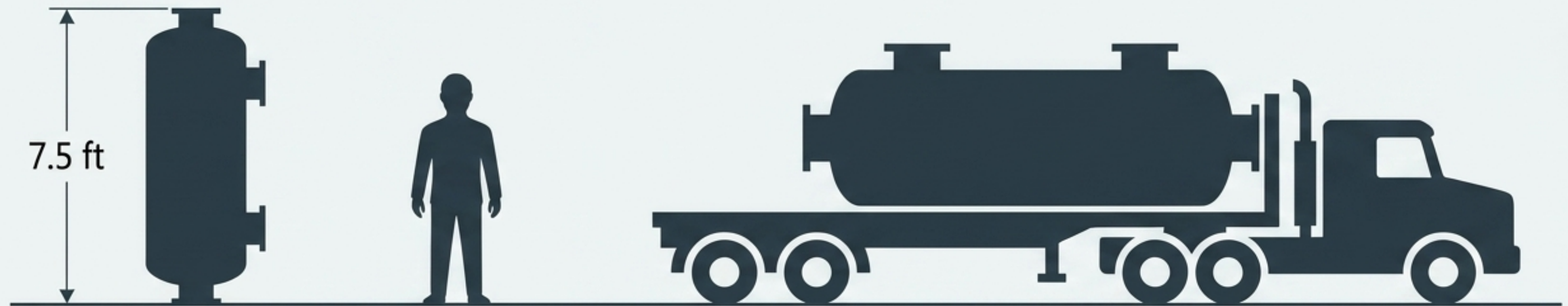


Reading Between the Lines: The 7.5-Foot Paradigm

Observation: Both Inlet Scrubbers (Sec 8.2.3.1) and Glycol Contactors (Sec 8.2.3.2) explicitly default to a standard 7 1/2-ft shell height.

The Implicit Insight: Why exactly 7.5 feet? The textbook does not explain this dimension's origin. It is the perfect equilibrium point where physics meets logistics:

-  **Physics:** Provides the absolute minimum vertical travel distance for liquid droplets to overcome upward vapor velocity and settle by gravity.
-  **Logistics:** Perfectly aligns with modular skid fabrication limits and standard highway transportation regulations without requiring oversize load permits.



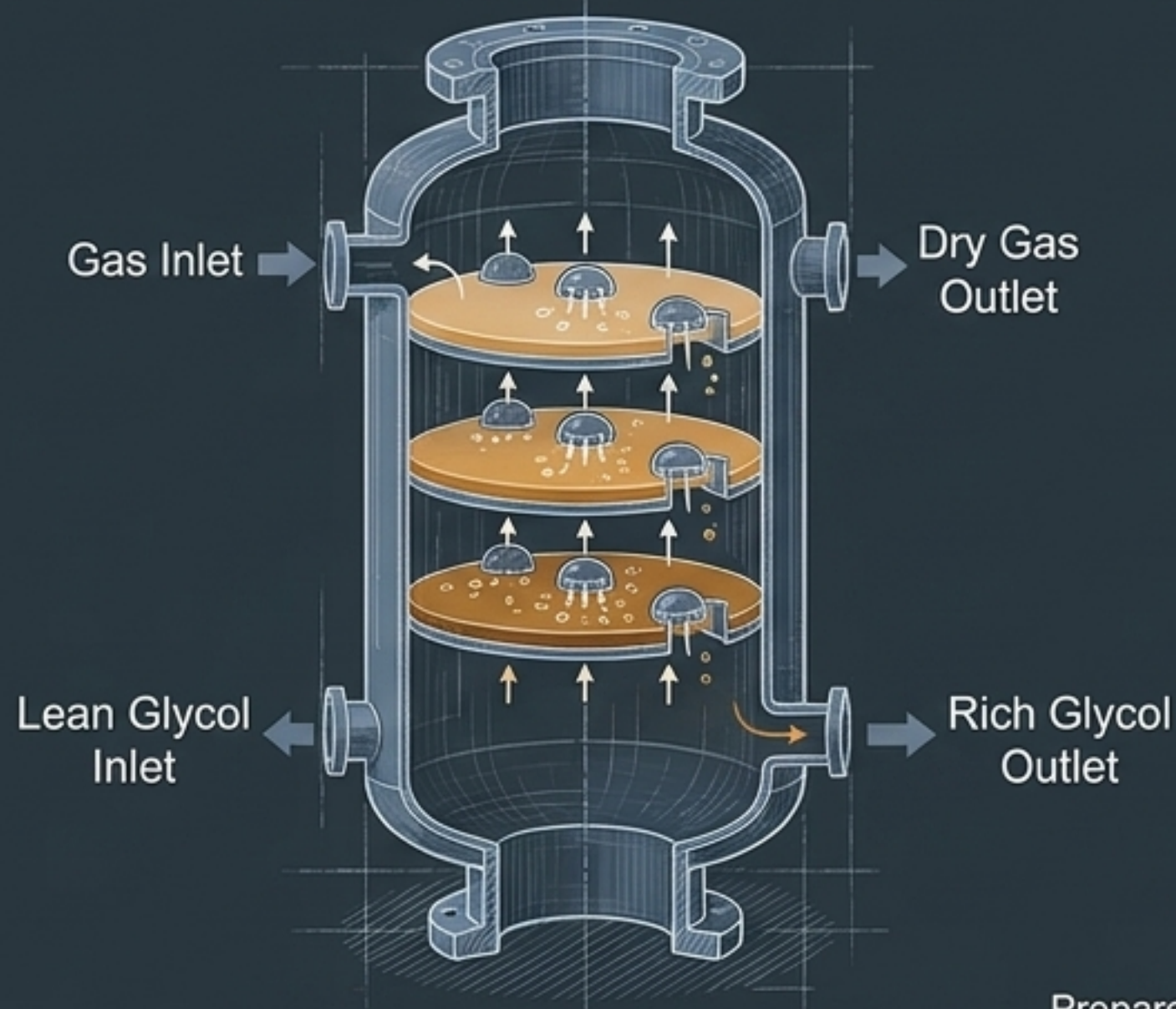
The Heart of the System: Glycol Contactors

The contactor is where dehydration physically occurs via mass transfer.

Standard Sizing Anchor: The minimum required diameter (OD) is determined entirely by the baseline capacity (q_s) of standard gas (0.7 SG at 100°F).

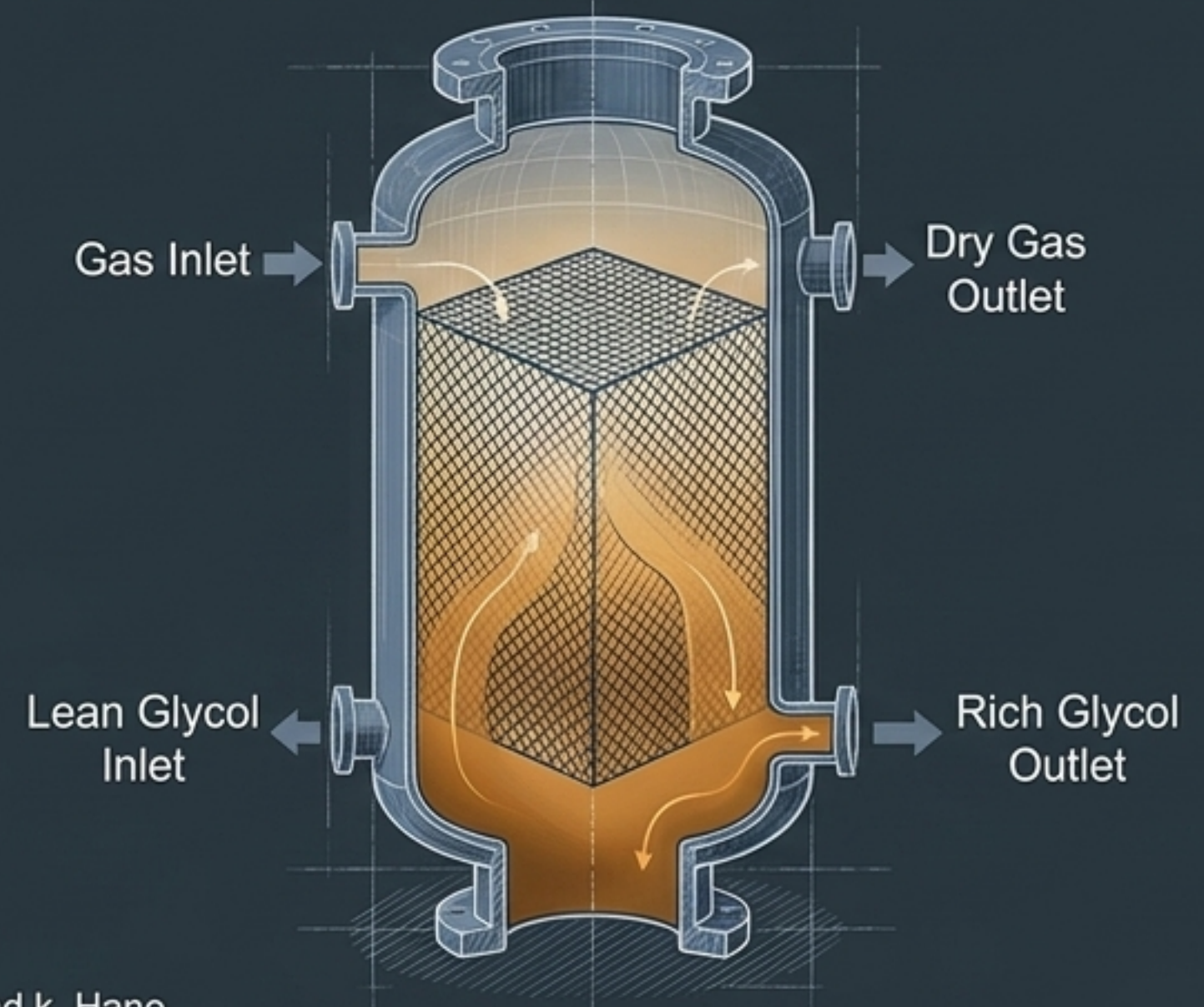
Trayed Contactors

Discrete staging. Gas pushes upward through discrete liquid layers on horizontal plates.



Packed Contactors

Continuous staging. Gas and liquid share an infinite, chaotic surface area through a structured media matrix.



Architecture Showdown: Trayed vs. Packed

Comparison Dimension	 Trayed	 Packed
Capacity Limit at 24-inch OD, 1000 psia	10.77 MMscfd	9.59 MMscfd
Maximum Published Vessel Sizing	Up to 72 inches OD (Handling over 122 MMscfd)	Limited to 24 inches OD in the baseline charts

Key Insight

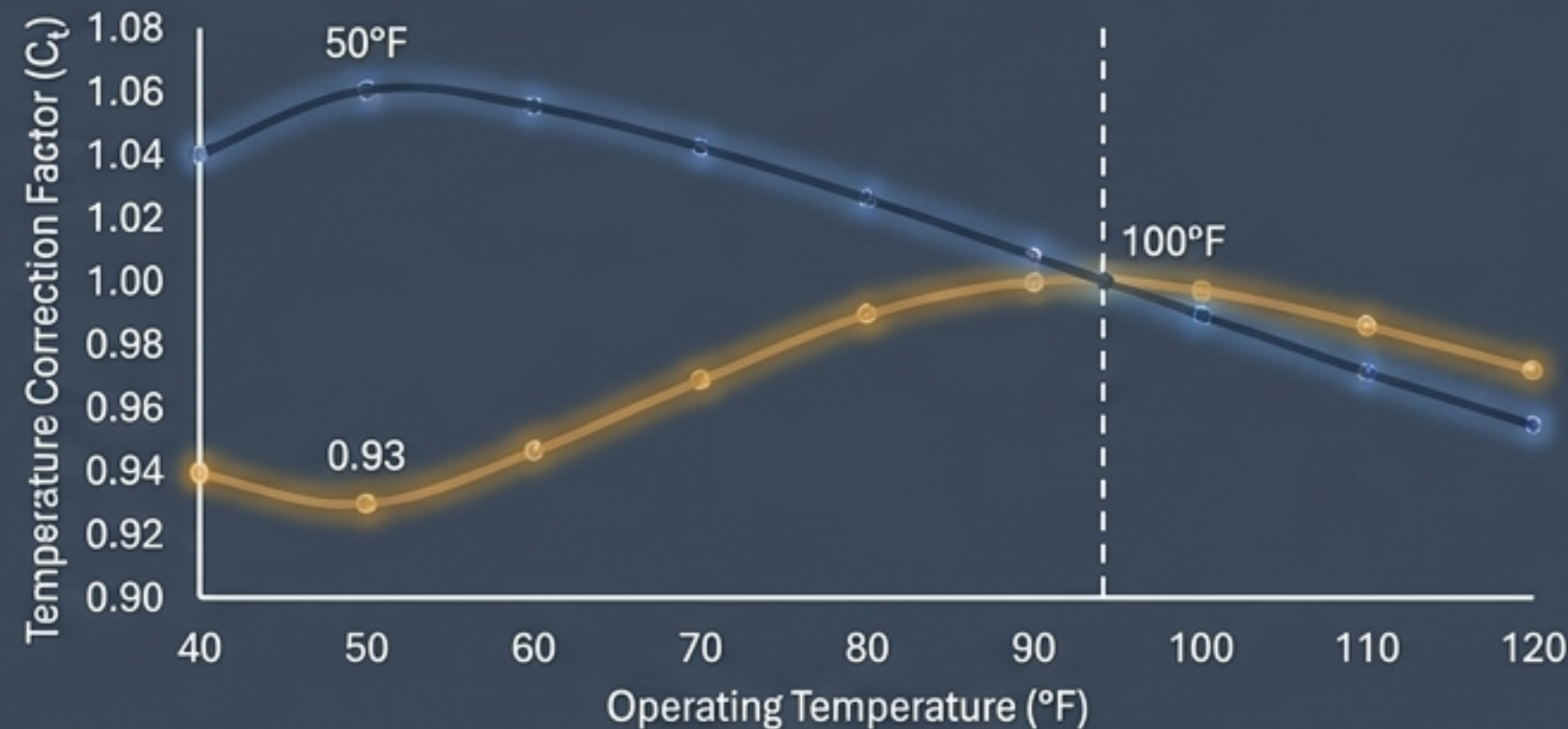
Trayed towers dominate at massive industrial scales due to predictable liquid distribution. Packed towers excel in **smaller, weight-sensitive footprints** but **suffer from channeling risk** at larger **diameters**.

The Thermodynamics of Correction: Eq 8.1 Divergence

$$q_s = \frac{q}{C_t \times C_g}$$

Correction Factors for Operating Temperature and Gas-Specific Gravity

Modifier Divergence Chart



Trayed Systems:

As temperature drops to 50°F, $C_t = 1.06$. The math rewards trayed systems at lower temperatures.

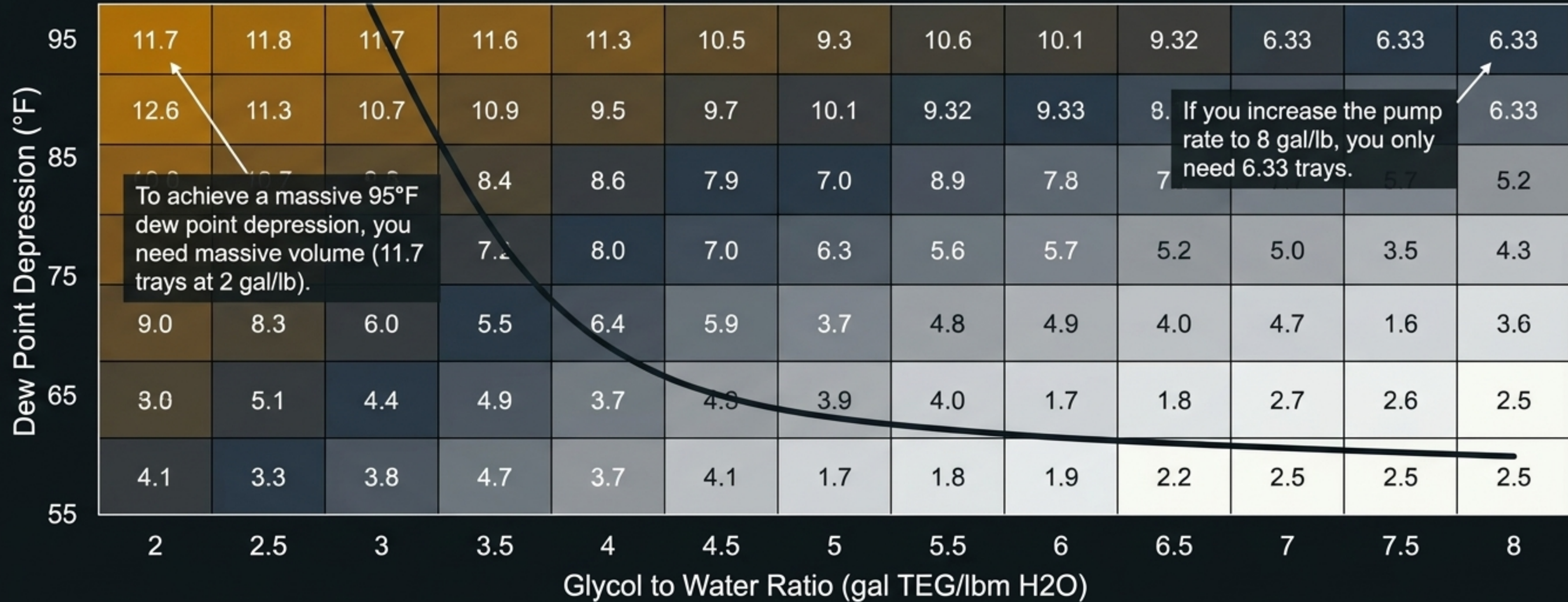
Packed Systems:

As temperature drops to 50°F, $C_t = 0.93$. The math penalizes packed systems.

Why?

Lower temperatures increase glycol viscosity. Bubble caps force gas through thick liquid forcefully (reward). Packed media relies on gravity trickling, which viscous, cold glycol resists (penalty).

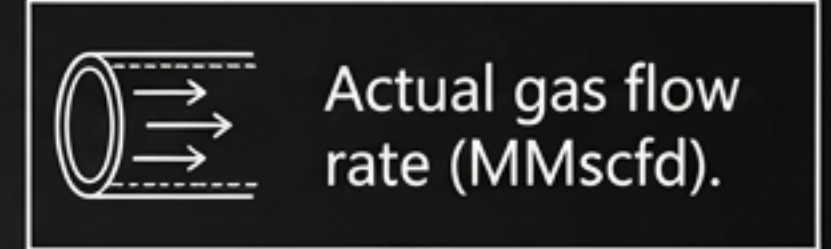
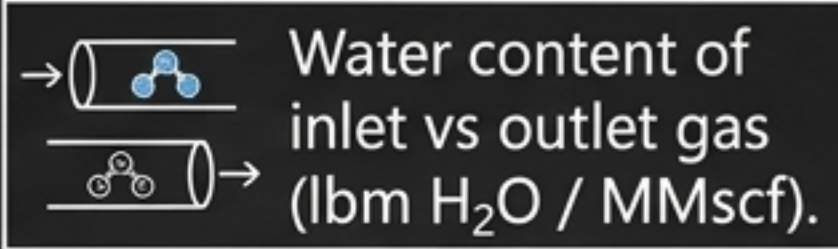
Internals Sizing: The Diminishing Returns of Glycol



The Asymptotic Limit

The curve flattens out. Pumping more than 5 or 6 gal/lbm of water yields rapidly diminishing physical returns. You cannot out-pump a tower that lacks sufficient vertical contact stages.

The Mass Balance: Quantifying Water Removal



$$W_r = (C_{wi} - C_{wo}) \times \frac{q}{24}$$



Target water to be removed (lbm/hr).

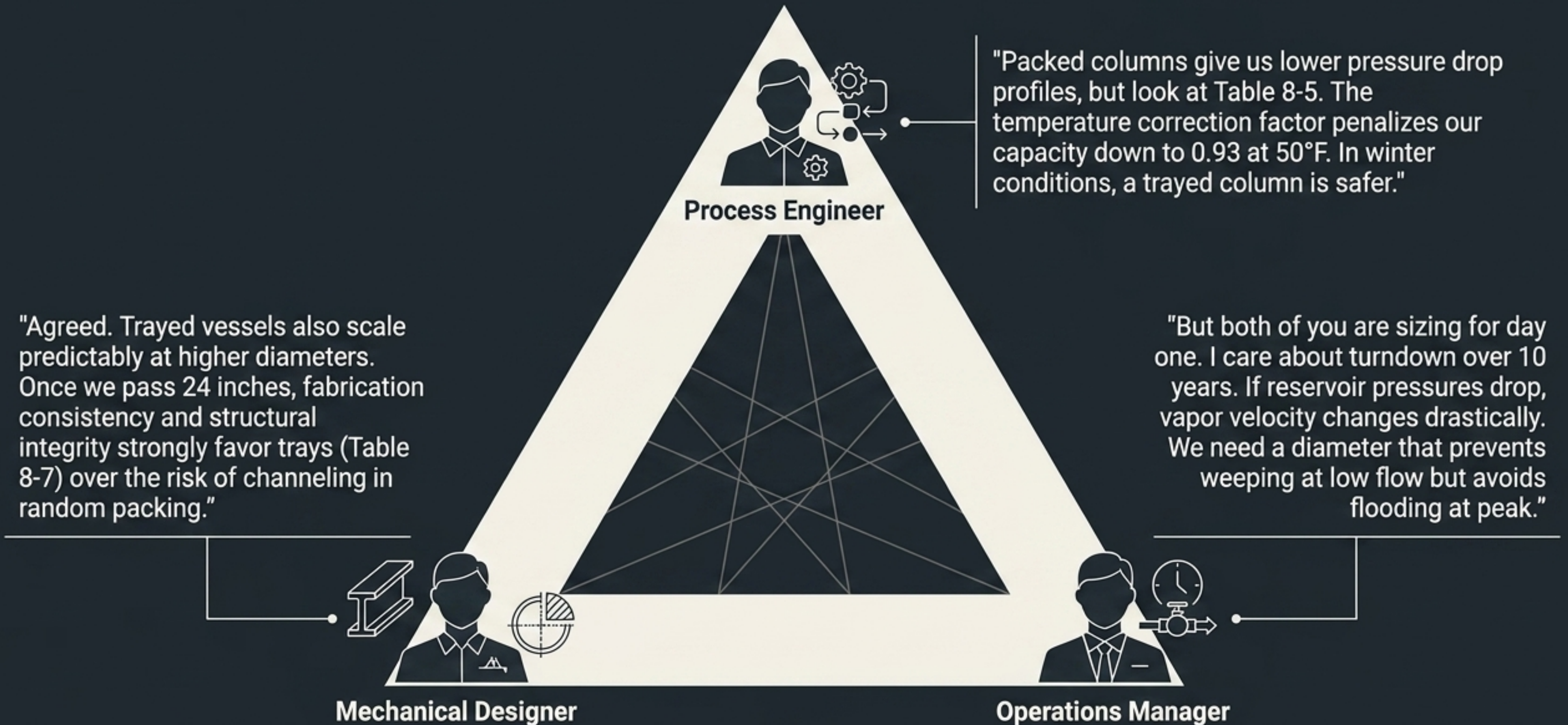


Conversion from daily capacity to hourly physical water removal.

Takeaway

This is the ultimate proof of design. If the scrubber and contactor boundaries are correctly sized, W_r matches the thermal capacity of the regeneration reboiler.

Expert Triangulation: Navigating the Trade-Offs

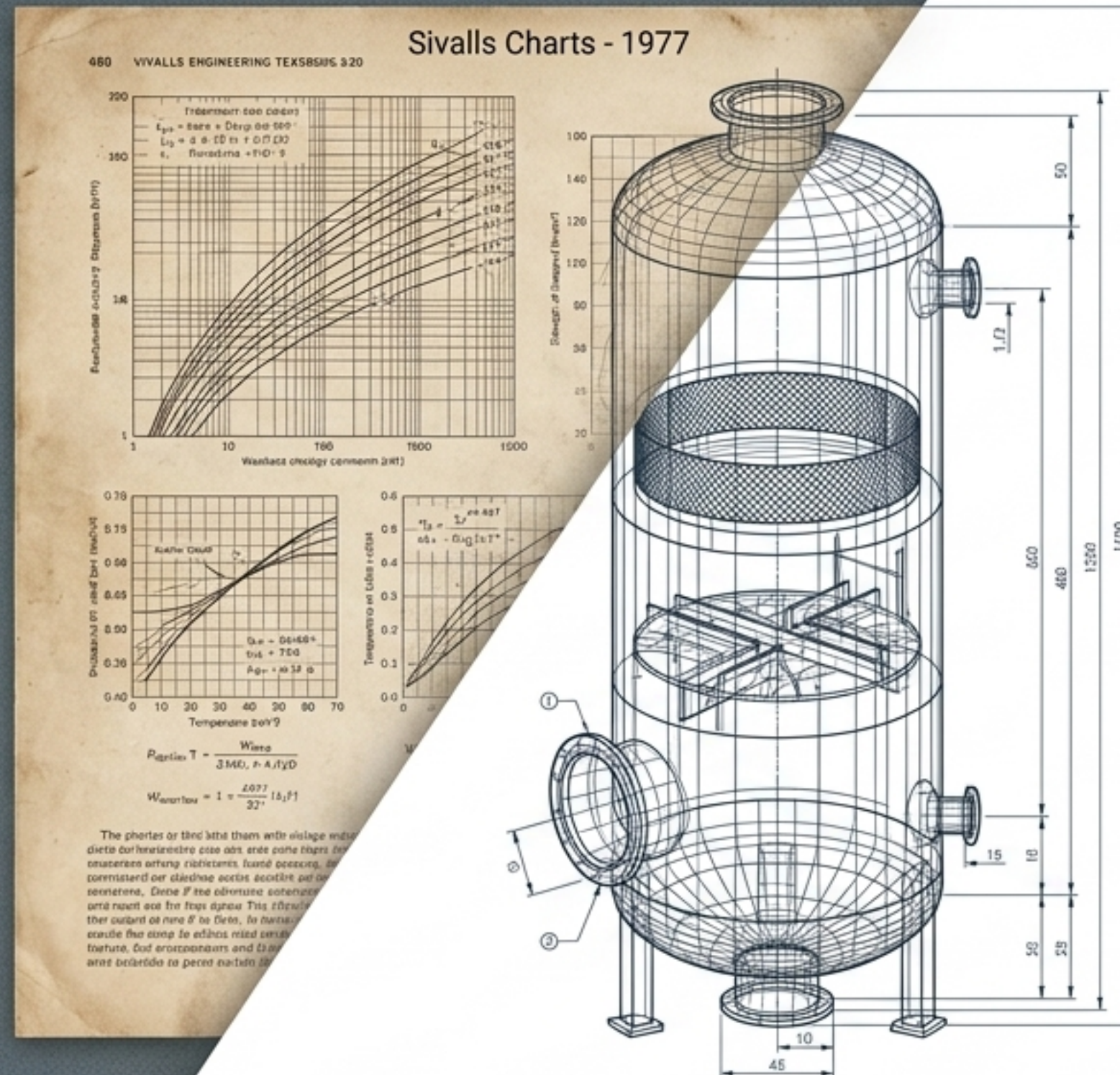


Prepared by Engineer Sajjad k. Hano

Meta-Analysis: The Sivalls Paradox

The Paradox:

Why do modern dehydration engineers armed with HYSYS and ProMax still rely on empirical charts published by Sivalls in 1977?

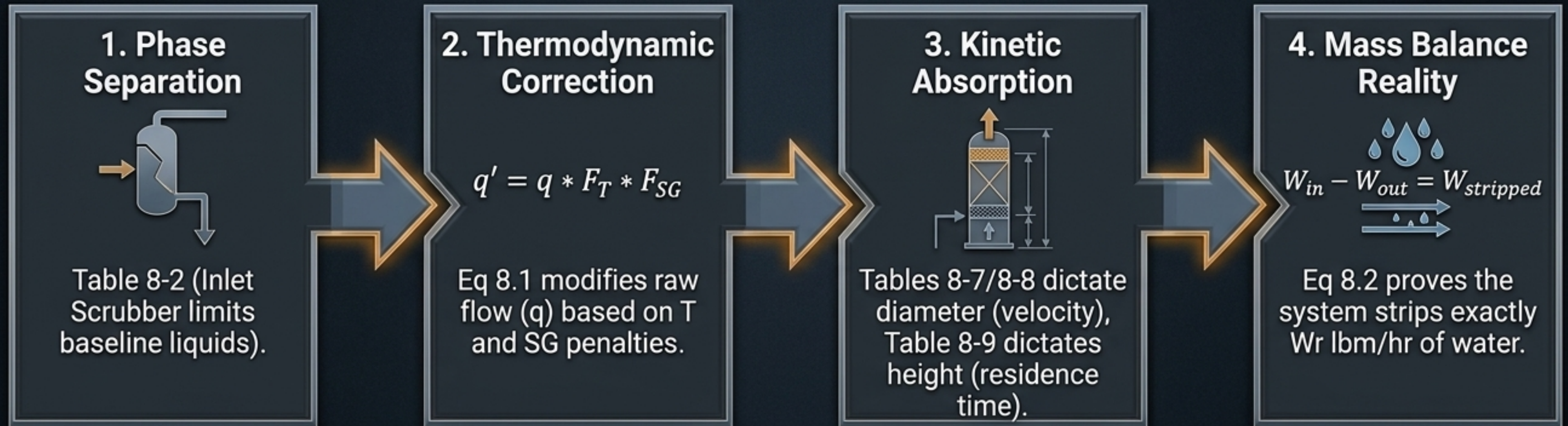


The Meta-Insight:

True physical principles are immutable. Advanced dynamic simulation software is powerful but risks becoming a “black box,” detaching the user from physical reality.

The Sivalls charts act as the ultimate cognitive anchor. They force the engineer to physically visualize the thermodynamic friction between velocity, pressure, and surface area before writing a line of code.

Synthesis: The Dehydration Ecosystem Architecture



Insight: A failure in Step 1 (Scrubber carryover) instantly invalidates the math in Step 3, destroying the equilibrium. The system is a single living mathematical organism 

The Operational Imperative

1. Vessels are Boundaries, not Buckets

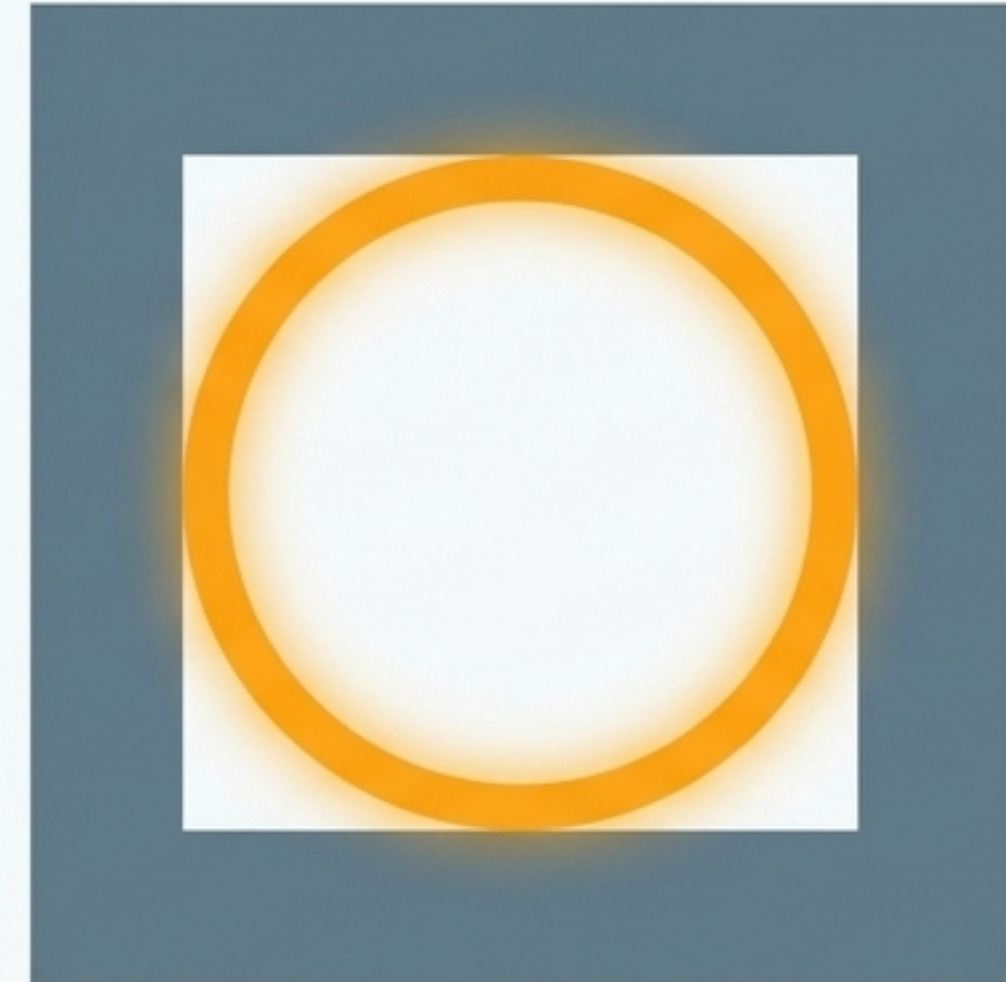
Sizing dictates kinetic velocity and residence time; it controls the physics of separation, not just volumetric capacity.

2. Thermodynamics Demand Trade-offs

Temperature, pressure, and Specific Gravity penalize or reward internals differently. Always calculate qs before selecting hardware.

3. Precision Equals Longevity

Proper application of empirical limits prevents catastrophic carryover, channeling, and foaming, safeguarding CAPEX and product purity.



Master the physical limits before you rely on the software.